

ANVESHAKHUB ENTERPRISE ERP

STANDALONE QA & END-TO-END TESTING MANUAL

Exhaustive Step-by-Step Testing Manual for All 52 Frontend Screens, Real Account Onboarding, Clean Database Policy, Security Hardening Verification & Standardized Bug Reporting

Testing Parameter	Specification / Value
Target System	AnveshakHub Enterprise Resource Planning (ERP)
Manual Version	v1.0 — Standalone QA & E2E Testing Manual (Exhaustive Unabridged)
Testing Policy	Clean Database Policy — Dynamic Real Account Provisioning (Zero Hardcoded Credentials)
Coverage Scope	52 Frontend Screens 10 Operational Modules 171 REST APIs 71 Unit Tests Passing
Target Audience	QA Lead, Software Testers, Security Reviewers, Full-Stack Engineers & Product Managers

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SECTION 1 — QA PREPARATION & CLEAN DATABASE SETUP

Clean Database Policy:

AnveshakHub ERP enforces a strict **Clean Database Policy**. No hardcoded, dummy, or fake test accounts exist in the production database. All testing must be conducted by creating real corporate entities, onboarding real employee profiles, and registering real client accounts dynamically.

Database Reset & Purge Procedures:

To reset the environment to a completely clean database state before a test cycle:

1. Execute ``npx ts-node -P apps/api/tsconfig.json prisma/seed/clean_db.ts`` in the project repository.
2. The system purges all transactional records, projects, deliverables, attendance logs, and documents.
3. Only the environment-configured Bootstrap Admin account (``BOOTSTRAP_ADMIN_EMAIL``) is initialized.

SECTION 2 — DYNAMIC REAL ACCOUNT PROVISIONING WORKFLOWS

Role / Account Type	Provisioning Workflow & Screen	How Account Credentials Are Created
System Admin	Environment Configuration / Seed	Configured via <code>`BOOTSTRAP_ADMIN_EMAIL`</code> and <code>`BOOTSTRAP_ADMIN_PASSWORD`</code> in <code>`.env`</code> .
HR Manager	Admin Onboarding Wizard (<code>`/hr/onboard`</code>)	System Admin logs in and creates HR user with work email and initial password.
Project Manager	HR Onboarding Wizard (<code>`/hr/onboard`</code>)	HR Manager creates PM user account with professional role <code>`PM`</code> and work email.
Employee / Intern	HR Onboarding Wizard (<code>`/hr/onboard`</code>)	HR Manager onboards employee, inputs real employee code, department, and work email.
Client (ORG_USER)	Public Org Registration (<code>`/register`</code>)	Client submits real org details on <code>`/register`</code> . Admin approves org on <code>`/admin/approvals`</code> .

SECTION 4 — DETAILED SCREEN-BY-SCREEN STEP-BY-STEP PROCEDURES (52 SCREENS)

Below is the complete step-by-step test execution manual for ALL 52 frontend screens in ``apps/web/app/``. Each screen includes target route, required role, step-by-step actions, form input details, expected UI behavior, and expected API responses:

TC-SCR-001: Landing Page (``/``)

Required Role: Public

UI Components: Navbar, Hero section, Feature cards, Footer links

Step-by-Step Test Procedure: 1. Open browser to ``/``. 2. Observe top navbar logo, hero banner, feature highlights, and call-to-action buttons. 3. Click 'Login' or 'Register' button.

Expected Result & API Behavior: Page renders cleanly without visual glitches. Clicking 'Login' redirects user to ``/login`` route.

TC-SCR-002: User Account Activation (``/activate``)

Required Role: Public / User

UI Components: Token validation indicator, Password input, Confirm password input, Submit button

Step-by-Step Test Procedure: 1. Receive activation link with query parameter ``?token=VALID_TOKEN``. 2. Open ``/activate?token=VALID_TOKEN`` in browser. 3. Enter desired account password in form. 4. Click 'Activate Account' button.

Expected Result & API Behavior: API sends ``POST /api/v1/auth/activate``. Account status updates to ACTIVE. User is redirected to ``/login`` with success alert.

TC-SCR-003: Admin Organization Approvals (``/admin/approvals``)

Required Role: ADMIN

UI Components: Pending applications list, Organization tax ID badge, Approve button, Reject modal

Step-by-Step Test Procedure: 1. Log in as System Admin. 2. Navigate to ``/admin/approvals``. 3. View pending organization registration applications. 4. Click 'Approve' button on target organization card. 5. Alternatively, click 'Reject' and provide rejection reason.

Expected Result & API Behavior: API sends ``PATCH /api/v1/organizations/:id/decision``. Organization status updates to ``APPROVED``. Registration token generated for client.

TC-SCR-004: Admin Organization Directory (``/admin/organizations``)

Required Role: ADMIN

UI Components: Search input, Organization datatable, Active/Inactive status toggle, Edit modal

Step-by-Step Test Procedure: 1. Navigate to ``/admin/organizations``. 2. Use search input to filter organizations by legal name. 3. Click on organization status toggle (Active/Inactive). 4. Click 'Edit Organization' to modify legal metadata.

Expected Result & API Behavior: API sends ``GET /api/v1/organizations`` and ``PATCH /api/v1/organizations/:id``. Status toggle updates database record immediately.

TC-SCR-005: Admin Problem Statement Review (``/admin/problem-statements``)

Required Role: ADMIN

UI Components: Filter tabs (Pending, Accepted, Rejected), Category dropdown, Status update modal

Step-by-Step Test Procedure: 1. Navigate to ``/admin/problem-statements``. 2. View submitted problem statements from client organizations. 3. Select statement, assign technical category/domain, and update review status to ``ACCEPTED`` or ``REJECTED``.

Expected Result & API Behavior: API sends `PATCH /api/v1/projects/problem-statements/:id`. Status updates in DB; notification sent to submitting client org.

TC-SCR-006: Admin System Monitor Telemetry (`/admin/system-monitor`)

Required Role: ADMIN

UI Components: PIN input modal, Infrastructure health gauges, Memory usage chart, Failed email queue table, Clear failed emails button

Step-by-Step Test Procedure: 1. Navigate to `/admin/system-monitor`. 2. System prompts for Argon2id PIN verification. 3. Enter admin PIN. 4. Observe real-time server memory, CPU load, active sessions, failed email queue, and system alerts.

Expected Result & API Behavior: API sends `POST /api/v1/system-monitor/verify-pin` (timing-safe check) and `GET /api/v1/system-monitor/metrics`. Infrastructure telemetry updates dynamically.

TC-SCR-007: Admin Workshop Management (`/admin/workshops`)

Required Role: ADMIN

UI Components: Workshop catalog list, Create workshop modal, Form inputs (Title, Date, Capacity), Save button

Step-by-Step Test Procedure: 1. Navigate to `/admin/workshops`. 2. Click 'Create Workshop' button. 3. Input workshop title, description, instructor, start date, and seat capacity. 4. Click 'Save Workshop'.

Expected Result & API Behavior: API sends `POST /api/v1/workshops`. Workshop is created and listed under upcoming workshops catalog.

TC-SCR-008: Global Document Repository (`/documents`)

Required Role: All Authenticated

UI Components: Folder tree view, Document datatable, Entity filter dropdown, Upload document modal

Step-by-Step Test Procedure: 1. Navigate to `/documents`. 2. Use folder tree navigation to browse system folders. 3. Filter documents by entity type (Project, Employee, Organization). 4. Click 'Upload Document' button.

Expected Result & API Behavior: API sends `GET /api/v1/documents`. Documents and folders render according to user's authorization boundary.

TC-SCR-009: Document Detail & Download Gate (`/documents/[id]`)

Required Role: All Authorized

UI Components: Metadata card, SHA-256 checksum badge, Scan status pill (PENDING, CLEAN, INFECTED), Download button

Step-by-Step Test Procedure: 1. Click on a specific document from `/documents`. 2. View document metadata (filename, MIME type, file size, SHA-256 checksum, scanStatus). 3. Click 'Download File' button.

Expected Result & API Behavior: API sends `GET /api/v1/documents/:id/download-url`. If `scanStatus === 'CLEAN'`, signed S3 URL is returned; if `PENDING` or `INFECTED`, API returns `HTTP 403 Forbidden`.

TC-SCR-010: Employee Document Repository (`/documents/employees/[id]`)

Required Role: HR / Admin / Self

UI Components: Pre-created folder grid, File list, Upload dropzone

Step-by-Step Test Procedure: 1. Navigate to employee document repository. 2. Browse pre-initialized folders (Identity, Education, Employment, Certifications). 3. Upload document into specific folder.

Expected Result & API Behavior: API sends `POST /api/v1/documents/upload-url` with S3 key prefix `quarantine/`. Document is registered with `scanStatus: PENDING`.

TC-SCR-011: Organization Document Repository (`/documents/organizations/[id]`)

Required Role: Org User / Admin

UI Components: Organization folder tree, File list, Download button

Step-by-Step Test Procedure: 1. Navigate to organization document repository. 2. View legal registration contracts and NDA attachments.

Expected Result & API Behavior: API checks organization membership boundary (`organizationId`). Access denied if user belongs to another organization.

TC-SCR-012: Project Deliverables Repository (`/documents/projects/[id]`)

Required Role: Project Team / PM

UI Components: Deliverables attachment list, Version history table, Presigned download links

Step-by-Step Test Procedure: 1. Navigate to project document repository. 2. Review milestone deliverables and requirement attachments.

Expected Result & API Behavior: Project deliverable attachments listed. Access restricted to assigned project team members and client org.

TC-SCR-013: Daily Attendance Portal (`/employee/attendance`)

Required Role: Employee / Workforce

UI Components: Current time digital clock, Clock In button, Clock Out button, Today's status badge, Monthly attendance calendar

Step-by-Step Test Procedure: 1. Log in as Employee. 2. Navigate to `/employee/attendance`. 3. Click 'Clock In' button. 4. Browser requests Geolocation permission. 5. Verify clock-in timestamp and status update to `PRESENT`. 6. At shift end, click 'Clock Out'.

Expected Result & API Behavior: API sends `POST /api/v1/attendance/clock-in` with browser IP and location coordinates. Double clock-in throws `400 Bad Request`.

TC-SCR-014: Employee Self-Service Dashboard (`/employee/dashboard`)

Required Role: Employee / Workforce

UI Components: Welcome banner, Clock-in quick action button, Leave balance progress bars, Assigned tasks list

Step-by-Step Test Procedure: 1. Log in as Employee. 2. View `/employee/dashboard`. 3. Check attendance status card, annual leave balance summary, and assigned active project tasks.

Expected Result & API Behavior: API sends `GET /api/v1/employee/dashboard-summary`. Summary cards render real-time database state.

TC-SCR-015: Employee Personal Documents (`/employee/documents`)

Required Role: Employee / Workforce

UI Components: Personal file table, Upload certificate button, Document version history

Step-by-Step Test Procedure: 1. Open `/employee/documents`. 2. View personal files (ID proof, certificates, contracts). 3. Click 'Upload Certificate'.

Expected Result & API Behavior: Uploaded document enters `quarantine/` storage path with `scanStatus: PENDING`.

TC-SCR-016: Employee Leave Portal (`/employee/leave`)

Required Role: Employee / Workforce

UI Components: Leave balance cards, Leave request form, Application history datatable, Cancel request button

Step-by-Step Test Procedure: 1. Open `/employee/leave`. 2. View annual leave balances (Casual, Medical, Annual). 3. Fill leave application form (Select type, Start Date, End Date, Reason). 4. Click 'Submit Application'.

Expected Result & API Behavior: API sends `POST /api/v1/leave/requests`. Request is created with status `PENDING`. Annual balance checked for overflow.

TC-SCR-017: Employee Profile Settings (`/employee/profile`)

Required Role: Employee / Workforce

UI Components: Profile avatar, Personal details form, Contact info inputs, Save button

Step-by-Step Test Procedure: 1. Open `/employee/profile`. 2. View employment details (Code, Designation, Department, Date Joined). 3. Update contact phone number and emergency contact details. 4. Click 'Save Profile'.

Expected Result & API Behavior: API sends `PATCH /api/v1/users/profile`. Profile updated in DB; audit event logged (`UPDATE\_PROFILE`).

TC-SCR-018: Forgot Password Recovery (`/forgot-password`)

Required Role: Public

UI Components: Email input field, Send reset link button, Confirmation alert box

Step-by-Step Test Procedure: 1. Open `/forgot-password`. 2. Input registered work email address. 3. Click 'Send Reset Link'.

Expected Result & API Behavior: API sends `POST /api/v1/auth/forgot-password`. Single-use `PasswordResetToken` generated with 1-hour expiry. Reset email dispatched.

TC-SCR-019: HR Management Dashboard (`/hr`)

Required Role: HR Manager / Admin

UI Components: Headcount metric cards, Department distribution pie chart, Pending leave requests table, Quick onboard button

Step-by-Step Test Procedure: 1. Log in as HR Manager. 2. Open `/hr`. 3. Review total employee headcount gauge, today's attendance rate, and pending leave applications.

Expected Result & API Behavior: API sends `GET /api/v1/hr/dashboard-metrics`. Displays organization-wide HR statistics.

TC-SCR-020: HR Attendance Summary & History (`/hr/attendance`)

Required Role: HR Manager / Admin

UI Components: Date range picker, Department filter dropdown, Attendance datatable, Export CSV button

Step-by-Step Test Procedure: 1. Open `/hr/attendance`. 2. Filter attendance history by date range and department. 3. Observe employee clock-in times, IP addresses, and total work hours.

Expected Result & API Behavior: API sends `GET /api/v1/attendance/summary` scoped strictly by requesting HR's `organizationId` (M-01 compliant).

TC-SCR-021: HR Detailed Employee Management (`/hr/employees/[id]`)

Required Role: HR Manager / Admin

UI Components: Employee header card, Employment details form, Department dropdown, Status toggle, Sensitive audit indicator

Step-by-Step Test Procedure: 1. Open employee profile from HR directory. 2. View complete profile, employment status, salary details, and document folder. 3. Edit designation or department assignment. 4. Click 'Save Changes'.

Expected Result & API Behavior: API sends `GET /api/v1/hr/employees/:id` and `PATCH /api/v1/hr/employees/:id`. Sensitive profile read audit event logged (M-05).

TC-SCR-022: HR Leave Review & Approval (`/hr/leave`)

Required Role: HR Manager / Admin

UI Components: Pending leave request table, Employee balance indicator, Approve button, Reject modal

Step-by-Step Test Procedure: 1. Open `/hr/leave`. 2. Review pending employee leave requests. 3. Click 'Approve' or 'Reject' button. 4. If rejecting, provide rejection note.

Expected Result & API Behavior: API sends `PATCH /api/v1/hr/leave/requests/:id/approve`. Status updates to `APPROVED`; leave quota deducted from `LeaveBalance` table.

TC-SCR-023: HR Employee Onboarding Wizard (`/hr/onboard`)

Required Role: HR Manager / Admin

UI Components: 3-Step wizard stepper, Form inputs (Code, Email, Department, Salary), Complete onboarding button

Step-by-Step Test Procedure: 1. Open `/hr/onboard`. 2. Step 1: Input employee code, first name, last name, work email, and phone. 3. Step 2: Select department, designation, and employment type. 4. Step 3: Set initial salary and start date. 5. Click 'Complete Onboarding'.

Expected Result & API Behavior: API sends `POST /api/v1/hr/onboard`. `User` and `Employee` records created in single DB transaction; invitation email dispatched.

TC-SCR-024: Industry Client Portal Home (`/industry`)

Required Role: ORG_USER (Client)

UI Components: Client organization header, Active projects grid, Problem statements status summary, Support ticket quick link

Step-by-Step Test Procedure: 1. Log in as Client `ORG\_USER`. 2. Open `/industry`. 3. View active client projects, submitted problem statements, and support tickets.

Expected Result & API Behavior: API sends `GET /api/v1/industry/dashboard`. Scope restricted to client's `organizationId`.

TC-SCR-025: Industry Support Contact (`/industry/contact`)

Required Role: ORG_USER (Client)

UI Components: Support message form, Category dropdown, Attachment input, Submit button

Step-by-Step Test Procedure: 1. Open `/industry/contact`. 2. Fill support message form. 3. Click 'Send Message'.

Expected Result & API Behavior: API sends `POST /api/v1/industry/support-ticket`. Support ticket created for admin review.

TC-SCR-026: Industry Project Deliverables Portal (`/industry/deliverables`)

Required Role: ORG_USER (Client)

UI Components: Deliverable list table, Status badge (Submitted, Accepted, Revision Requested), Download button, Acceptance modal

Step-by-Step Test Procedure: 1. Open `/industry/deliverables`. 2. View submitted deliverables for client projects. 3. Click 'Review Deliverable' to accept or request revision. 4. Download deliverable document.

Expected Result & API Behavior: API sends `PATCH /api/v1/projects/deliverables/:id/review`. Deliverable file download gated by `scanStatus: CLEAN`.

TC-SCR-027: Industry Documents (`/industry/documents`)

Required Role: ORG_USER (Client)

UI Components: Document list table, Entity filter, Download link

Step-by-Step Test Procedure: 1. Open `/industry/documents`. 2. Browse organization contracts, NDAs, and project attachments.

Expected Result & API Behavior: API checks client organization boundary. Cross-tenant document access returns `403 Forbidden`.

TC-SCR-028: Industry Meetings Schedule (`/industry/meetings`)

Required Role: ORG_USER (Client)

UI Components: Meeting calendar, Meeting list card, Videoconference join link

Step-by-Step Test Procedure: 1. Open `/industry/meetings`. 2. View upcoming videoconference review meetings with project managers.

Expected Result & API Behavior: API sends `GET /api/v1/industry/meetings`. Meeting schedule rendered cleanly.

TC-SCR-029: Industry Problem Statements List (`/industry/problem-statements`)

Required Role: ORG_USER (Client)

UI Components: Problem statements table, Status pills (Pending, Under Review, Accepted), Submit new statement button

Step-by-Step Test Procedure: 1. Open `/industry/problem-statements`. 2. View submitted problem statements and review status.

Expected Result & API Behavior: API sends `GET /api/v1/industry/problem-statements`. Displays organization's problem statements.

TC-SCR-030: Industry Problem Statement Detail (`/industry/problem-statements/[id]`)

Required Role: ORG_USER (Client)

UI Components: Statement header, Description card, Review notes list, Linked project link

Step-by-Step Test Procedure: 1. Click on problem statement from list. 2. View detailed problem description, assigned review committee, and linked enterprise project.

Expected Result & API Behavior: API sends `GET /api/v1/industry/problem-statements/:id`. Shows review notes and linked project.

TC-SCR-031: New Problem Statement Submission (`/industry/problem-statements/new`)

Required Role: ORG_USER (Client)

UI Components: Multi-field form (Title, Description, Outcomes, Budget), Document dropzone, Submit button

Step-by-Step Test Procedure: 1. Open `/industry/problem-statements/new`. 2. Fill title, business domain, problem description, expected outcomes, and budget range. 3. Attach specification document. 4. Click 'Submit Problem Statement'.

Expected Result & API Behavior: API sends `POST /api/v1/industry/problem-statements`. Statement created with status `PENDING\_REVIEW`.

TC-SCR-032: Industry Organization Profile (`/industry/profile`)

Required Role: ORG_USER (Client)

UI Components: Org legal details card, Tax ID display, Authorized contacts list, Edit contact modal

Step-by-Step Test Procedure: 1. Open `/industry/profile`. 2. View client organization legal details, tax ID, business domain, and authorized contacts.

Expected Result & API Behavior: API sends `GET /api/v1/industry/profile`. Displays organization legal metadata.

TC-SCR-033: Industry Projects List (`/industry/projects`)

Required Role: ORG_USER (Client)

UI Components: Project cards grid, Status badges (In Progress, Completed), Progress bar

Step-by-Step Test Procedure: 1. Open `/industry/projects`. 2. View active projects commissioned by client organization.

Expected Result & API Behavior: API sends `GET /api/v1/industry/projects`. Scoped strictly to client organization.

TC-SCR-034: Industry Project Detail View (`/industry/projects/[id]`)

Required Role: ORG_USER (Client)

UI Components: Project overview card, Milestone progress timeline, Sanitized team count badge (names hidden)

Step-by-Step Test Procedure: 1. Click on project from list. 2. View sanitized project overview, milestone progress, and deliverable review timeline.

Expected Result & API Behavior: API sends `GET /api/v1/industry/projects/:id`. Sanitizes internal employee identities for client privacy.

TC-SCR-035: Industry Query Support Tickets (`/industry/queries`)

Required Role: ORG_USER (Client)

UI Components: Support ticket datatable, Ticket status pills (Open, In Progress, Resolved), New ticket button

Step-by-Step Test Procedure: 1. Open `/industry/queries`. 2. View open and resolved technical support query tickets. 3. Click 'New Support Ticket'.

Expected Result & API Behavior: API sends `GET /api/v1/industry/queries`. Listed query tickets.

TC-SCR-036: Industry Query Response Thread (`/industry/queries/[id]`)

Required Role: ORG_USER (Client)

UI Components: Message thread chat box, Reply textarea, Send reply button

Step-by-Step Test Procedure: 1. Click on query ticket. 2. Read conversation history between client and support team. 3. Type reply message and click 'Send Reply'.

Expected Result & API Behavior: API sends `POST /api/v1/industry/queries/:id/messages`. Response message appended to thread.

TC-SCR-037: Secure Login Portal (`/login`)

Required Role: Public

UI Components: Email input, Password input, Remember me checkbox, Sign In button, Forgot password link

Step-by-Step Test Procedure: 1. Open `/login`. 2. Enter registered work email address and password. 3. Click 'Sign In' button.

Expected Result & API Behavior: API sends `POST /api/v1/auth/login`. Sets `HttpOnly; Secure; SameSite=Lax` access token cookie. User redirected to primary role dashboard.

TC-SCR-038: User Notifications Inbox (`/notifications`)

Required Role: All Authenticated

UI Components: Notification list items, Unread badge counter, Mark as read button, Filter tabs

Step-by-Step Test Procedure: 1. Open `/notifications`. 2. View list of system notifications (Leave status, Task assignment, Deliverable review). 3. Click 'Mark as Read'.

Expected Result & API Behavior: API sends `GET /api/v1/notifications` and `PATCH /api/v1/notifications/:id/read`. Unread count updates instantly.

TC-SCR-039: Organizations Directory (`/organizations`)

Required Role: All Authenticated

UI Components: Search input, Organization cards grid, Business domain tags

Step-by-Step Test Procedure: 1. Open `/organizations`. 2. Search approved client organizations and business partners.

Expected Result & API Behavior: API sends `GET /api/v1/organizations`. Returns active approved organizations.

TC-SCR-040: Platform Privacy Policy (`/privacy`)

Required Role: Public

UI Components: Legal document text formatting, Print page button

Step-by-Step Test Procedure: 1. Open `/privacy`. 2. Verify platform data privacy policy and GDPR/compliance disclosures.

Expected Result & API Behavior: Static text page renders cleanly.

TC-SCR-041: User Account Settings (`/profile`)

Required Role: All Authenticated

UI Components: Account summary card, Change password form, Active sessions table

Step-by-Step Test Procedure: 1. Open `/profile`. 2. View user account details and security settings. 3. Click 'Change Password'. 4. Input current password and new password. 5. Click 'Update Password'.

Expected Result & API Behavior: API sends `POST /api/v1/auth/change-password`. Password hashed via Argon2id; audit event logged (`UPDATE\_PASSWORD`).

TC-SCR-042: Internal Projects Directory (`/projects`)

Required Role: Workforce / PM

UI Components: Projects datatable, Category filter, Status dropdown, Create project button

Step-by-Step Test Procedure: 1. Log in as PM or Employee. 2. Open `/projects`. 3. View enterprise projects where user is an active member. 4. Click 'Create Project' (PM/Admin only).

Expected Result & API Behavior: API sends `GET /api/v1/projects`. Returns projects assigned to user or organization.

TC-SCR-043: Internal Project Management Workspace (`/projects/[id]`)

Required Role: Project Team / PM

UI Components: Project tab bar (Overview, Team, Milestones, Deliverables, Resources), Add milestone modal, Assign member dropdown

Step-by-Step Test Procedure: 1. Open project detail workspace. 2. View project milestones, deliverables, assigned team members, and resource requirements. 3. Click 'Add Milestone' or 'Assign Member'.

Expected Result & API Behavior: API sends `GET /api/v1/projects/:id` and `POST /api/v1/projects/:id/members`. Manages project execution.

TC-SCR-044: Client Organization Registration (`/register`)

Required Role: Public

UI Components: Registration form fields (Org Name, Org Number, Tax ID, Admin Email), Submit registration button

Step-by-Step Test Procedure: 1. Open `/register`. 2. Input company legal name, registration number, tax ID, business domain, and primary admin contact details. 3. Click 'Submit Registration'.

Expected Result & API Behavior: API sends `POST /api/v1/organizations/registration`. Organization created with status `PENDING\_APPROVAL`.

TC-SCR-045: Organization Registration Status (`/registration-status`)

Required Role: Public

UI Components: Org number search input, Check status button, Application status result card

Step-by-Step Test Procedure: 1. Open `/registration-status`. 2. Input organization registration number. 3. Click 'Check Status'.

Expected Result & API Behavior: API sends `GET /api/v1/organizations/status/:regNumber`. Displays approval status (PENDING_APPROVAL, APPROVED, REJECTED).

TC-SCR-046: Password Reset Form (`/reset-password`)

Required Role: Public

UI Components: New password input, Confirm password input, Reset password button

Step-by-Step Test Procedure: 1. Open `/reset-password?token=VALID\_RESET\_TOKEN` from email link. 2. Input new account password and confirmation. 3. Click 'Reset Password'.

Expected Result & API Behavior: API sends `POST /api/v1/auth/reset-password`. Token validated for single-use status; password updated via Argon2id (H-02 compliant).

TC-SCR-047: Global ERP Search (`/search`)

Required Role: All Authenticated

UI Components: Global search input, Categorized results tabs (Projects, Employees, Documents), Search result cards

Step-by-Step Test Procedure: 1. Open `/search` or use header search bar. 2. Type search query (e.g. 'Project Alpha' or 'John Doe'). 3. Observe search results categorized by Projects, Employees, and Documents.

Expected Result & API Behavior: API sends `GET /api/v1/system/search?q=QUERY`. Results filtered strictly by user's tenant authorization boundary.

TC-SCR-048: Enterprise Help Desk & Support (`/support`)

Required Role: All Authenticated

UI Components: User guide links, FAQ accordion, Support ticket form

Step-by-Step Test Procedure: 1. Open `/support`. 2. View system user guide, FAQ accordion, and submit help desk ticket.

Expected Result & API Behavior: Help desk documentation rendered; support ticket submitted to enterprise admin.

TC-SCR-049: Terms of Service (`/terms`)

Required Role: Public

UI Components: Legal text formatting, Acceptable use policy sections

Step-by-Step Test Procedure: 1. Open `/terms`. 2. Verify platform terms of service and acceptable use policy.

Expected Result & API Behavior: Static text page renders cleanly.

TC-SCR-050: Unauthorized Access Banner (`/unauthorized`)

Required Role: All Authenticated

UI Components: 403 Forbidden graphic, Unauthorized access alert, Return to dashboard button

Step-by-Step Test Procedure: 1. Attempt accessing a route restricted to another role (e.g., Employee accessing `/admin/system-monitor`). 2. System renders `/unauthorized` screen.

Expected Result & API Behavior: RoleGuard denies access; user redirected to `/unauthorized`.

TC-SCR-051: Public Workshops Catalog (`/workshops`)

Required Role: Public / User

UI Components: Workshop grid cards, Date badges, Instructor info, Register button

Step-by-Step Test Procedure: 1. Open `/workshops`. 2. Browse available corporate workshops, dates, and seat availability. 3. Click 'Register for Workshop'.

Expected Result & API Behavior: API sends `GET /api/v1/workshops`. Lists available workshops.

TC-SCR-052: Workshop Detail & Registration (`/workshops/[id]`)

Required Role: Public / User

UI Components: Curriculum list, Instructor card, Participant registration form, Submit registration button

Step-by-Step Test Procedure: 1. Open workshop detail page. 2. View complete curriculum breakdown, instructor bio, and registration form. 3. Submit workshop registration.

Expected Result & API Behavior: API sends `POST /api/v1/workshops/:id/register`. Registration recorded for participant.

SECTION 6 — SECURITY TESTING MANUAL (SAFE TEST SUITE)

QA Security Verification Matrix for audit findings A through G:

Audit Finding	Test Action / Vector	Expected Security Defense	Status
C-01 JWT Cookie	Check `window.localStorage` in DevTools.	Zero tokens stored in localStorage; HttpOnly cookie used.	PASS
H-01 Revocation	Send API request using token after logout.	Backend returns 401 Unauthorized; token revoked.	PASS
H-02 Reset Bypass	Attempt reset with raw userId in payload.	Backend rejects request; single-use token required.	PASS
H-03 Path Traversal	Request download for `../../etc/passwd`.	`validateStorageKey` rejects key with 400 Bad Request.	PASS
M-01 Attendance Isolation	HR Org A requests attendance of Employee Org B.	Backend filters query by Org A; Org B records omitted.	PASS
M-02 Doc Quarantine	Request download for file with `scanStatus: PENDING`.	Backend rejects download with 403 Forbidden gating.	PASS
Section F BOLA	ORG_USER requests `/api/v1/documents/overview/organizations`.	Backend restricts list to user's assigned organization.	PASS

SECTION 7 — STANDARDIZED BUG REPORTING PROTOCOL

Format for reporting issues: Bug ID, Affected Route/Screen, Action/Button Triggered, Step-by-Step Reproduction, Expected vs. Actual Result, Severity (P0 Critical, P1 High, P2 Medium, P3 Low).

SECTION 8 — TESTING TEAM EXECUTION MATRIX (5-TESTER ALLOCATION)

Tester Role	Assigned Modules & Routes	Primary Focus Areas
Tester 1 (Auth & Admin)	`/login`, `/register`, `/forgot-password`, `/admin/*`	HttpOnly cookie auth, PIN verification, org approvals, system monitor.
Tester 2 (HR & Attendance)	`/hr/*`, `/employee/attendance`, `/employee/profile`	Employee onboarding, clock-in/out, attendance history, profile audits.
Tester 3 (Leave & Workspace)	`/employee/leave`, `/hr/leave`, `/profile`	Leave applications, annual quota calculations, HR approvals/cancellations.
Tester 4 (Projects & Industry)	`/projects/*`, `/industry/*`	Problem statements, milestone progress, team allocation, client portal.
Tester 5 (Documents & Security)	`/documents/*`, `/notifications`, `/search`	Presigned upload URLs, quarantine pathing, scan gating, global search.